MediAssist AI

Intelligent Patient Pre-Consultation and Doctor Decision Support System

B.Tech Fourth-Year Minor Project - Evaluation and Startup Potential

Overall concept assessment: 9.2 / 10
A strong AI-oriented project concept with high academic demonstration potential, provided that the system is designed as a doctor-assistance tool and not an autonomous prescriber.

Prepared from the project idea and evaluation discussion

1. Project Idea (User's Question)

We plan to develop a completely AI-based web application for healthcare. When a patient arrives, the patient will describe symptoms, medical history and other relevant details to the system. The AI will analyze the case, prepare a structured summary and suggest relevant clinical considerations and possible medicines for the doctor to review. The case will then be presented to the doctor, who will verify the information and make the final clinical decision and prescription. The goal is to reduce the time doctors spend handling each patient. We also want to evaluate this idea as a B.Tech fourth-year minor project and explore whether it could become a product sold to doctors.

2. Overall Evaluation

This idea has excellent potential for a B.Tech fourth-year minor project and promising commercial potential as a healthcare startup. The recommended positioning is a **doctor-assistance system**, not a system that independently diagnoses patients or prescribes medicines.

Evaluation area	Score	Assessment
Overall project concept	**9.2/10**	Excellent potential
Technical complexity	9.0/10	AI, NLP, backend, database and safety workflow
Innovation and usefulness	9.2/10	Can reduce documentation and intake burden
Academic demonstration	9.5/10	Strong if tested with measurable outcomes
Startup/market potential	8.5/10	Promising, but requires customer validation

These are qualitative concept-level assessments, not official university marks or a guarantee of commercial success.

3. Proposed System Workflow

1. Patient interaction

Patient describes symptoms, duration, medical history, allergies and current medicines through a web interface.

2. AI clinical intake engine

An LLM extracts relevant details, identifies missing information, structures the history and creates a case summary.

3. Safety and decision support

The system flags potential emergency warning signs, highlights allergies and possible contraindication concerns, and presents clinical considerations for the doctor.

4. Doctor dashboard

The doctor sees the structured summary, relevant history, alerts and supporting information.

5. Doctor review and prescription

The doctor examines the patient, corrects the summary if needed, accepts or rejects suggestions and independently makes the final clinical decision.

Core value proposition: reduce administrative workload and organize patient information accurately, while keeping clinical responsibility and final prescribing authority with the doctor.

4. Possible B.Tech Evaluation Rubric (100 Marks)

Criterion	Maximum marks
Problem definition and objectives	10
System architecture and database design	10
Web application and user interface	10
AI/LLM integration and NLP	20
Clinical workflow and decision-support logic	15
Testing and performance evaluation	15
Security, privacy and safety	10
Documentation, presentation and viva	10
TOTAL	100

Indicative outcome by implementation quality:

- Basic implementation: 65–75/100
- Well-developed AI, database and testing: 80–90/100
- Excellent implementation with measured results, safety checks and a convincing demo: 90–95/100

Actual marks depend on the university's rubric and the quality of the submitted implementation.

5. Recommended Features

Intelligent patient interview

Natural-language symptom collection and relevant follow-up questions about onset, duration, severity, medical history and associated symptoms.

Automatic clinical case summary

Structured chief complaint, symptoms, duration, medical history, allergies, current medicines and unanswered questions.

Safety and emergency alerts

Flag predefined warning signs and direct the user toward urgent human assessment when appropriate; do not delay emergency care for a complete interview.

Evidence-grounded decision support

Show possible clinical considerations and cited guideline information. Any medicine-related suggestions should be restricted, clinically reviewed and checked for relevant contraindications.

Doctor dashboard and review

Allow the doctor to correct the summary, record the assessment and make the final prescription decision.

Performance analytics

Measure preparation time, summary completeness, fact accuracy, missing information, AI output quality and doctor editing effort.

6. Suggested Technology Stack

Layer	Suggested technology
Front end	React, or HTML/CSS/JavaScript

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Back end	Python with FastAPI or Django
AI and NLP	LLM API, structured outputs and prompt engineering
Clinical knowledge	Curated, cited medical guidelines and clinician-reviewed rules
Database	PostgreSQL
Authentication	Role-based patient and doctor accounts
Deployment	Cloud-hosted web application
Testing	Python tests, scenario-based evaluation and usability testing

A large language model does not need to be trained from scratch for the student prototype. A practical design combines an existing LLM with structured extraction, reviewed clinical rules and a carefully controlled workflow.

7. Scientific Evaluation Plan

Metric	What it measures
Summary completeness	Whether important patient facts were captured
Fact accuracy	Whether the AI preserved the patient's actual statements
Follow-up question relevance	Whether questions were useful and appropriate
Emergency-alert sensitivity	Whether predefined high-risk test scenarios were flagged
Preparation time	Time to prepare a case manually versus with the system
Doctor correction rate	How often the doctor had to correct AI output

Use synthetic or appropriately de-identified cases reviewed by qualified clinicians. Compare manual and AI-assisted preparation on the same test cases. Do not claim time savings or diagnostic accuracy until the measurements support those claims.

8. Startup Potential and Business Model

There is a plausible business opportunity if the product saves time, improves documentation, protects patient data and fits into the doctor's existing workflow. Willingness to pay should be validated with real clinics.

Customer segment	Possible offering
Individual doctors / small clinics	Patient intake, case summaries and a doctor dashboard
Multidoctor clinics	Multiple accounts, staff access, appointment workflows and centralized records
Hospitals / healthcare networks	Integration with existing systems, administrative controls and larger deployments

Illustrative pricing to test (not verified market rates):

• Starter: ■999–■1,999 per month
• Professional: ■2,999–■4,999 per month
• Multi-doctor clinic: ■7,500+ per month

Example only: 100 clinics paying ■2,000 per month would generate ■2 lakh monthly recurring revenue, or ■24 lakh annually, before expenses, taxes, acquisition costs and churn. This is a hypothetical scenario, not a forecast.

9. Major Risks and Safeguards

Clinical safety

An LLM can misunderstand symptoms or invent details. Important facts should be traceable to patient statements.

Patient privacy

Use informed consent, secure storage, access controls, encryption and audit logs.

Regulatory compliance

In India, software intended for medical purposes may fall under the Medical Devices Rules, 2017. Applicable requirements depend on intended use and risk classification; obtain specialist regulatory advice before clinical deployment.

Clinical validation

Use appropriate clinician review and testing before real-patient use.

Commercial adoption

Run a pilot with willing doctors and demonstrate benefit instead of assuming they will buy.

For the initial B.Tech demonstration, use synthetic patient cases and keep the system in a supervised, non-prescribing mode.

10. Final Recommendation

Verdict: 9.2/10 for the project concept. MediAssist AI is a strong candidate for a B.Tech fourth-year minor project and could provide a foundation for a startup. Its value will depend on a working prototype, clinical oversight, meaningful testing, privacy safeguards and a clear business model.

Recommended two-stage plan

Stage 1 - B.Tech minor project: Build patient interviews, structured case summaries, safety flags, a doctor dashboard and a measured evaluation.

Stage 2 - Startup product: Pilot with a few willing doctors, validate time savings, establish clinical and privacy safeguards, and then expand features and customers.

Do not attempt to build a complete AI doctor for every disease in the first version. A reliable system that prepares a high-quality patient case for a doctor is a stronger starting product than a broad system making unverified diagnostic and medicine recommendations.

Note: This document is a project-planning and academic evaluation aid, not medical advice, clinical validation, legal advice or a guarantee of marks or commercial success.